

How to Improve a Neural Network

If I have already trained ANW, how to improve it?

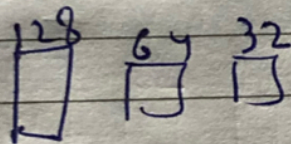
If $\rightarrow$ Acc $\rightarrow$ 90%.

$\rightarrow$ How to convert it to

92%.

1. Fine tune Neural Network Hyperparameters

- No. of hidden layers
- No. of Neurons per layer



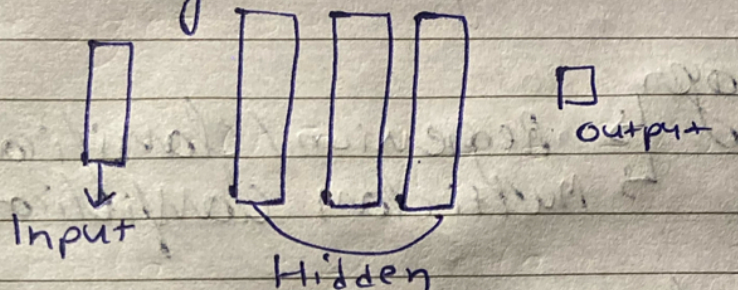
- Learning rate
- Optimizer
- Batch Size (Mini batch GD) — 32, 64, 128
- Activation function
- Epochs

2. By Solving problems:

- Vanishing & Exploding gradient
- Not enough data
- Slow training
- overfitting

Fine tuning Hyperparameters

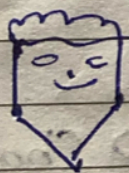
1. No. of hidden layer



Rather than taking one hidden layer & having more neuron in it, taking multiple hidden layers with fewer neuron

Why we need more layers?

Consider I am detecting face



Neural Network works on Representation layer such that first primitive layer will take, lines, edges middle layers add those lines and make shape & final layers add shapes & make face

So they work like this

Now, when How many to create until when my model doesn't perform overfitting, if it started I should stop.

It also uses transfer ^{learning} layer i.e., I can use layer, initial layer in any other model like for money face prediction

2. How many Neuron in layer

Input layer

↳ No. of Input columns

Output layer

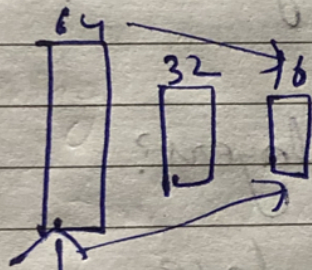
↳ Regression / Classification - 1

↳ Multiclass classification → No. of classes

Hidden layers

Pyramid Structure

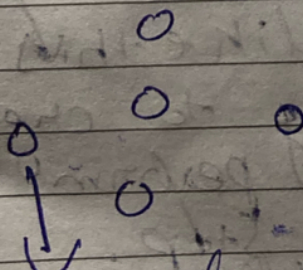
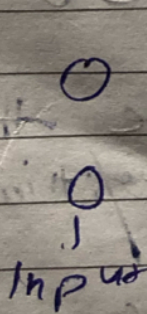
→ No much diff



Because it captures primitive features

So primitive features > Shapes > Faces

More Node Sufficient



We loose primitive layer because

No neuron to handle it

Always Start with high no. of neurons, if it makes problem reduce it

3. Batch Size

Batch GD, SGD, MBGD

↓
Batch Size = 1, No. of ram, 32, 64, ...

Batch Size

(8 to 32) Smaller → Slow training, generalization
↓
result on new data is good

(8 to 192) large → training fast, if results on data is good it is more efficient because it is faster.
So, use learning rate scheduler for good result.

↳ Depends on GPU RAM

Learning Scheduler

↓
Keep it low initially & making it more gradually this called warming up

40 Epochs → ?

use Early Stopping if we use it we comparing Accuracy with last epochs if it found No improvement it stops by own. this helps in fast computation & doesn't train for all epochs

Problems in Neural Networks

- Vanishing & Exploding gradients

- Weight init
- Activation fun.
- Batch Norm
- Gradient clipping

- Not enough data

- Transfer learning
- Unsupervised learning

- Slow training

- optimizers
- learning rate scheduler

- Overfitting

- L_1 & L_2 reg
- Dropout